Exam Date & Time: 21-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEMESTER B.TECH END SEMESTER EXAMINATION, NOVEMBER 2024

BASIC CLINICAL SCIENCE II [BME 3121]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Instructions to Candidates:
Answer ALL questions
Missing data may be suitably assumed

1A)	Discuss briefly the biological effects of radiation.	(4)
1B)	Explain the advantage and disadvantages of ultrasound.	(3)
1C)	Explain the various interactions of radiation with matter.	(3)
2A)	Explain the effect of main magnetic field on MR active nuclei inside the human body while placing in the magnetic field.	(5)
2B)	Define linear energy transfer, relative biological effectiveness and oxygen enhancement ratio.	(3)
2C)	Describe direct and indirect action of radiation.	(2)
3A)	Define theraupetic ratio with the help of a suitable diagram.	(3)
3B)	Describe the philosophy of three principles of radiation protection.	(3)
3C)	What is OCT? What is the principle of OCT?	(4)
4A)	What is Tonometry? Mention the different types of tonometry with an example for each. In which ocular disease Tonometry is important ?	(4)
4B)	Mention the tests done to assess visual acuity.	(2)

4C)	Describe the conditions necessary for EEG signals to be detected at the scalp. Why are large populations of pyramidal cells and synchronous activation required?	(4)
5A)	Define Peripheral Nerve Stimulation (PNS) and provide two examples of its application.	(4)
5B)	Explain the principle of Blood Oxygen Level-Dependent (BOLD) contrast in fMRI. How does it serve as an indirect measure of neuronal activity?	(3)
5C)	Describe the purpose and basic procedure of Nerve Conduction Studies (NCS). How do latency and amplitude measurements help in assessing nerve function?	(3)

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